

Introduction to Artificial Intelligence

Course syllabus, Fall 2026

Prof. Gilles Louppe

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This course is given by

- Theoretical lectures: Gilles Louppe
- Exercise sessions: Gilles Louppe, Fanny Bodart, Thomas Savary
- Programming projects: Fanny Bodart, Thomas Savary

Feel free to contact us at info8006@montefiore.ulg.ac.be or on [Discord](#) for help.

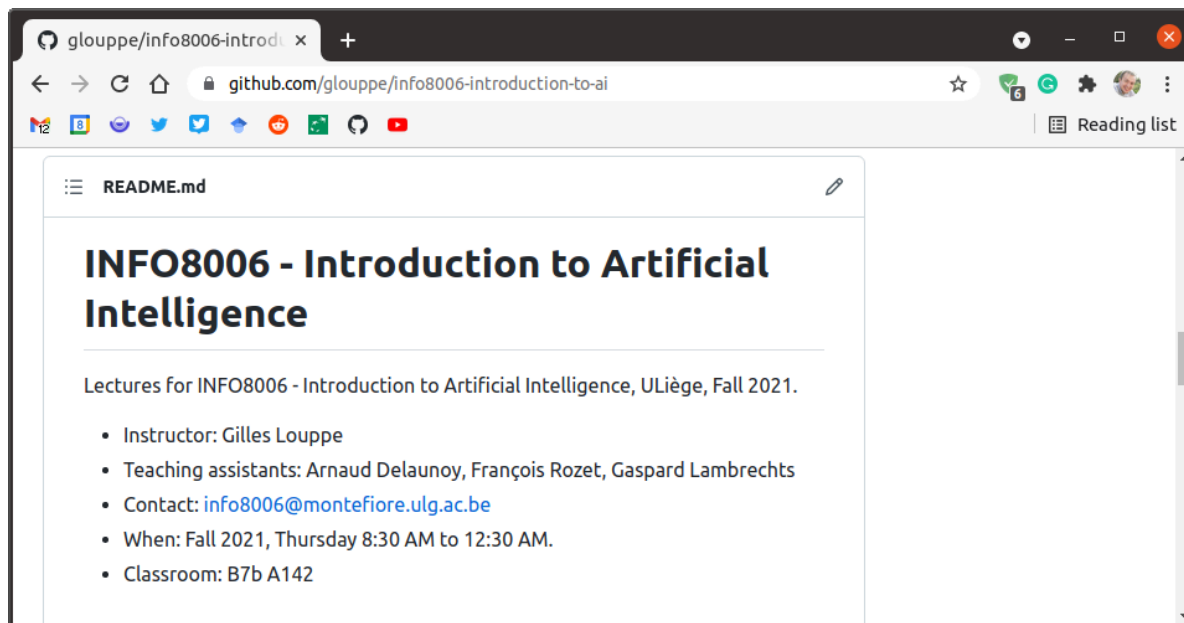


Materials

The schedule and slides are available at github.com/glouppe/info8006-introduction-to-ai.

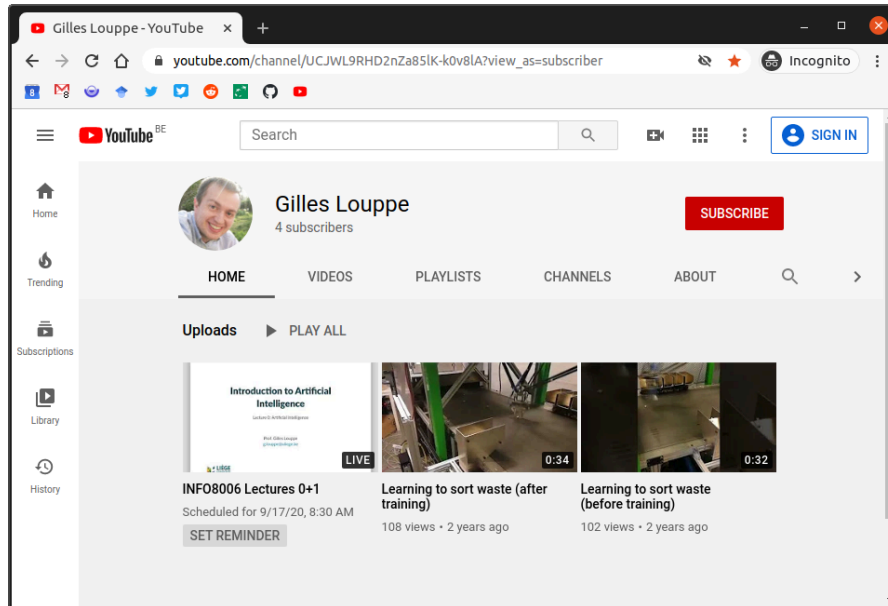
- In HTML and in PDFs.
- Minor updates up to the day before the lesson.

Official course description: [engagements pédagogiques](#).

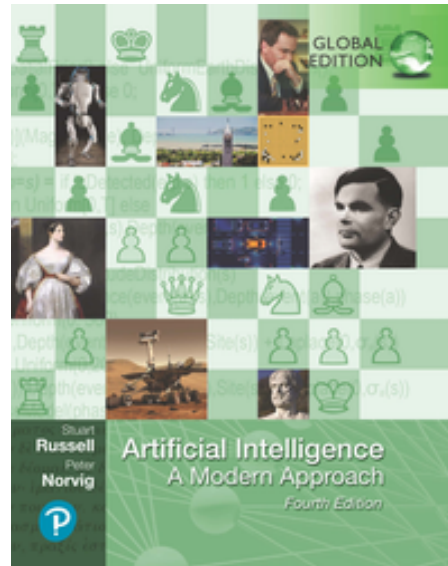


Videos

- Lectures from Fall 2024 are available on [MyUnicast](#) and on [YouTube](#).
- Videos from Fall 2020 are available on [YouTube](#). They are not up-to-date with the current course, but they may still be useful.



Textbook



The core content of this course is based on the following textbook:

*Stuart Russell, Peter Norvig. "Artificial Intelligence: A Modern Approach",
Fourth Edition, Global Edition.*

This textbook is **recommended**. It covers both the theory and the exercises.

CS188

- Lectures, exercises, and various other materials are partially adapted from [CS188 Introduction to Artificial Intelligence](#), from UC Berkeley.
- Cartoons that you will see in those slides were all originally made for CS188.



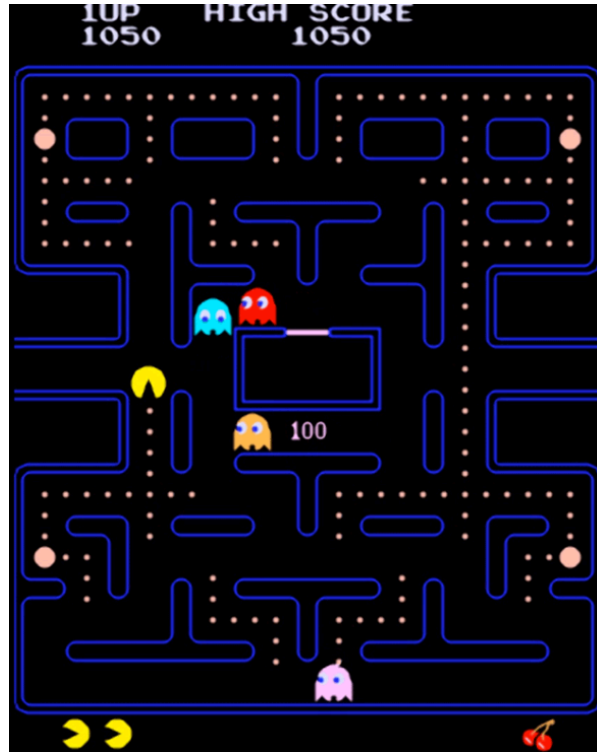
Exercise sessions

- Every week, after the lecture.
- You solve exercises alone or in small groups, and ask us when you are stuck.
- Solutions are provided for all exercises.

The written exam (70% of the grade) asks problems of the same kind, and you sit it alone, without documents. Reading a solution does not teach you to produce one. An AI assistant can solve the exercises for you, but it cannot sit the exam.

Programming projects

Implement an intelligent agent for playing **Pacman**. The project will be divided into three parts, with increasing levels of difficulty.



Evaluation

- Written exam (70%)
- Programming projects (30%)
 - Project 0: 0% (tutorial of September 24)
 - Project 1: 15%
 - Project 2: 15%
 - Programming projects are **mandatory** for presenting the exam.

A minimum of 8/20 is required for the written exam. Below that, the exam grade becomes the final grade.

Evaluation of the projects

- Projects are evaluated **automatically** on Gradescope.
- Preliminary tests are provided to help you debug your code.
- However, the final grade will be based on additional tests that are not provided.
 - It is part of the learning experience to design your own tests.
 - It is not because the public tests pass that your code is correct.
- The efficiency of your code will also be taken into account, beyond its correctness.
- Projects also include an **oral defense**: you answer questions on your own code, justify your design decisions, and may be asked to modify or extend your implementation.

Honor code

You may consult papers, books, online references, or publicly available implementations for ideas that you may want to adapt and incorporate into your projects, so long as you clearly cite your sources in your code and your writeup. *However, under no circumstances, may you base your project on someone else's implementation.*

Plagiarism is checked and sanctioned by a grade of 0. Cases of plagiarism will all be reported to the Faculty office.

Use of AI tools

AI assistants are **allowed**, without restriction.

The rule is simple: everything you submit, you must be able to explain. Submitting a solution you do not understand, because an AI assistant produced it, will be strongly penalized.

Let's start!